

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

QIAN QI

ABSTRACT. Finite projective pressures are organized by local zero-free complex atlases rather than a global complex logarithm. Bounded cylinder projections use convex-support face stratification instead of an impossible gradient blow-up at infinity. Phase coexistence is retained as a maximum of analytic phase charts, with phase-wise microscopic recovery and a lower-semicontinuous minimum rate. An explicit topology-upgrade lemma, genuine thin-shell conditioning, complete finite-dimensional duality hypotheses, and exact finite-centered local likelihoods yield a standalone stratified projective commutation theorem.

Platform identifier: HSBG-COMMUTE-v2. **Controlling revision:**
ROUND7-REFEREE-POSITIVE-CLOSURE.

CONTENTS

1. Round-seven reconstruction: stratified projective pressures, coexistence charts, and exact local likelihoods	1
1.1. Local zero-free pressure atlases	2
1.2. Finite-dimensional rates with bounded support and faces	2
1.3. Projective passage and topology upgrade	3
1.4. Conditioning on thin constraints	3
1.5. Contraction and primal-dual commutation	4
1.6. Exact local likelihoods and Gaussian tangents	4
1.7. Dynamic contractions	5
Scope and verification status	5

1. ROUND-SEVEN RECONSTRUCTION: STRATIFIED PROJECTIVE PRESSURES, COEXISTENCE CHARTS, AND EXACT LOCAL LIKELIHOODS

This section replaces the round-six controlling module. No global complex logarithm is assumed. Each real compact source set is covered by finitely

Date: August 31, 2026.

2020 *Mathematics Subject Classification.* Primary 60F10, 82C40; Secondary 60H10, 49L25.

Key words and phrases. kinetic large deviations, tangent laws, entropic rigidity, Girsanov theorem, Boltzmann equation.

many zero-free complex tubes. Bounded projections are treated by convex-support face stratification rather than by an impossible gradient blow-up at infinity. Nondifferentiable coexistence pressures are retained as maxima of phase charts and their lower bounds are obtained by phase-wise recovery.

1.1. Local zero-free pressure atlases. Let $\pi_m : E \rightarrow \mathbb{R}^{d_m}$ be a finite coordinate projection. For every compact real source set $K \subset \mathbb{R}^{d_m}$ assume the platform spectral or cluster theorem gives finitely many open sets $U_{m,j} \supset K_j$ covering K such that the finite-volume moment generating function has no zero on $U_{m,j} + iB(0, \eta_{m,j})$. Fix the logarithm by its real value at one point of K_j . The limiting phase pressure on that chart is analytic and denoted $Q_{m,j}$.

The physical pressure is the upper envelope

$$Q_m(\theta) = \max_{j \in \mathcal{J}_m(\theta)} Q_{m,j}(\theta).$$

At coexistence more than one phase may maximize and Q_m need not be differentiable.

Lemma 1.1 (Zero-free atlas normalization). *On overlaps, two logarithm branches differ by an integer multiple of $2\pi i/\mu_\varepsilon$ and agree on the real axis; hence their normalized finite-volume pressures and all real derivatives agree. Local uniform convergence on the tubes implies convergence of every finite real derivative on compact subsets. No single logarithm on all of $\mathbb{C}^{d_m}$ is used.*

Proof. The ratio of two nonvanishing analytic moment generating functions is one, so logarithm branches differ by a constant $2\pi i k$. Division by the speed and agreement at the real base point fix $k = 0$ for the normalized branch used on the overlap. Cauchy's formula gives derivative convergence. $\square$

1.2. Finite-dimensional rates with bounded support and faces. Let C_m be the closed convex hull of the support of $\pi_m(X)$. For a phase chart j define

$$I_{m,j}(x) = \sup_{\theta} \{\theta \cdot x - Q_{m,j}(\theta)\}.$$

Interior points of the phase mean image are reached by finite tilts. Boundary points of C_m are organized by exposed faces. If F is a face exposed by n_F , first send the normal source rn_F to infinity and then use a finite tangential tilt inside F . The tilted means remain bounded, as they must for bounded cylinders.

Theorem 1.2 (Face-stratified finite-dimensional LDP). *For every finite projection, the laws of $\pi_m(X_\varepsilon)$ satisfy a full good LDP with rate*

$$I_m(x) = \min_j I_{m,j}(x)$$

on phase-attainable points and its lower-semicontinuous extension to C_m . Interior lower bounds use finite analytic tilts. Boundary lower bounds use normal face concentration followed by tangential tilts. At coexistence the lower bound is the minimum over the individually recovered phase costs.

Proof. The local pressure atlas gives the exponential upper bound by finite covering of compact source sets. For an interior differentiability point of one phase, the exact normalized tilt and its pressure gap give the local lower bound. For an exposed face, the source rn_F concentrates the law exponentially on that face because all points outside have a uniform support-function gap. After that concentration, the tangential local pressure atlas gives the LDP inside the lower-dimensional affine span. Iterate over a flag of faces for non-smooth boundary points. Straszewicz approximation of compact convex sets and lower semicontinuity complete the boundary lower bound. If several phase pressures maximize, choose the phase-specific microscopic recovery supplied by the platform theorem; summing or mixing phase packets changes no exponential cost. Thus global differentiability is unnecessary. $\square$

1.3. Projective passage and topology upgrade. Let E_{proj} carry the initial topology generated by the maps π_m and let E be the desired weighted path topology. Assume the projections separate points and agree with the E topology on every compact rate sublevel.

Lemma 1.3 (Exponential-tightness topology upgrade). *If the laws are exponentially tight in E and satisfy the projective LDP in E_{proj} , then they satisfy the same LDP in E . The rate is*

$$I(x) = \sup_m I_m(\pi_m x).$$

Proof. For a closed E set, intersect with an exponentially good compact set. On that compact set the two topologies coincide, so the projective upper bound applies. The compact complement has arbitrarily large exponential cost. An E -open neighborhood of x contains a projective cylinder neighborhood after restriction to a compact rate sublevel; the projective lower bound then gives the E lower bound. Let the compact level increase. $\square$

Theorem 1.4 (Stratified projective full LDP). *Suppose every finite projection satisfies Theorem 1.2 and the platform laws are exponentially tight in E . Then the full laws satisfy a good LDP in E with the projective rate above. The theorem permits bounded cylinder coordinates, local complex charts, phase coexistence, and nondifferentiable pressure envelopes.*

Proof. Dawson–Gartner gives the LDP in the initial projective topology. Apply Lemma 1.3. Goodness follows from exponential tightness and the projective compactness of rate sublevels. $\square$

1.4. Conditioning on thin constraints. Let $C : E \rightarrow \mathbb{R}^r$ be a continuous finite coordinate constraint. A source-uniform coefficient theorem means that, at every regular constrained phase, exact finite saddles exist and the shrinking shell probability equals $e^{o(\mu_\varepsilon)}$ uniformly on bounded pulled-back source charts. For hard spheres this is B1; for a Sinai roof/homology shell it is A2.

Theorem 1.5 (Conditional projective LDP). *Under the source-uniform coefficient theorem, conditioning on the exact lattice coordinate and the admissible shrinking continuous shell yields the good rate*

$$I^a(x) = \begin{cases} I(x) - \inf_{Cy=a} I(y), & Cx = a, \\ +\infty, & Cx \neq a. \end{cases}$$

The conclusion is obtained from numerator and denominator shell asymptotics; it is not inferred from the unconditioned LDP alone.

Proof. Apply the coefficient theorem after every finite source tilt. The shell factor is subexponential and cancels between numerator and denominator, so the finite conditional pressures are the constrained envelopes. Their face-stratified lower bounds pass through the projective system. Exponential tightness on the constraint surface gives the full conditional LDP. $\square$

1.5. Contraction and primal–dual commutation. Let $T : E \rightarrow F$ be continuous. At the primal level both conditioning then contraction and contraction of the constrained rate give

$$J^a(y) = \inf\{I(x) - I_a : Cx = a, Tx = y\}.$$

For dual equality assume T and C are continuous linear maps on the finite projection and that the projected rate is proper, lower semicontinuous, convex, with a relative-interior feasible point.

Theorem 1.6 (No-gap commutation). *Under the stated finite-dimensional constraint qualification, the conjugate of the constrained contracted rate equals the constrained pressure evaluated on $T^*\eta$. At coexistence the equality is applied phase by phase and the lower semicontinuous envelope is taken afterwards. For nonlinear contractions only the primal contraction and the full C2 Frechet chain rule are asserted.*

Proof. The primal identity is the associativity of infima. On a finite linear projection, Fenchel–Rockafellar applies because the relative-interior point is a continuity point of the indicator of $Cx = a$. Phase-wise application avoids pretending that a nondifferentiable pressure envelope has one global Hessian. Projective passage follows by monotone enlargement of the finite source families. $\square$

1.6. Exact local likelihoods and Gaussian tangents. Fix one real regular source Θ and a zero-free complex neighborhood from Lemma 1.1. Put

$$m_{\varepsilon, \Theta} = DQ_{\varepsilon}(\Theta), \quad Z_{\varepsilon} = \sqrt{\mu_{\varepsilon}}(X_{\varepsilon} - m_{\varepsilon, \Theta}).$$

For fixed h define

$$L_{\varepsilon}(h) = \exp \left\{ h \cdot Z_{\varepsilon} - \mu_{\varepsilon} \left[Q_{\varepsilon}(\Theta + h/\sqrt{\mu_{\varepsilon}}) - Q_{\varepsilon}(\Theta) - \mu_{\varepsilon}^{-1/2} DQ_{\varepsilon}(\Theta)h \right] \right\}.$$

Theorem 1.7 (Exact finite-centred local likelihood). *The variable $L_\varepsilon(h)$ is exactly the Radon–Nikodym density of the local tilted law and has expectation one. If the finite pressure Hessians converge, then*

$$\log L_\varepsilon(h) = h \cdot Z_\varepsilon - \frac{1}{2} h^T \Sigma_\Theta h + o_P(1).$$

Process versions follow whenever the platform-specific B3 or A4 tightness theorem is available.

Proof. Substitute the definition of Z_ε ; the finite mean terms cancel algebraically, leaving the exact exponential-family likelihood. Taylor’s theorem inside the fixed zero-free chart gives the quadratic expansion. No entire pressure is needed. $\square$

1.7. Dynamic contractions. If a proved good path action is additive and has compact sublevels, define

$$V_t F(x) = \sup_{\gamma_0=x} \{F(\gamma_t) - I_{[0,t]}(\gamma)\}.$$

Theorem 1.8 (Rate–pressure–semigroup commutation). *For each platform satisfying the preceding model interfaces, contraction, conditioning, local likelihood tangents, and the Lax–Oleinik semigroup commute on their common finite source charts. Phase coexistence is represented by a minimum of phase actions and a maximum of phase pressures; the dynamic value selects the optimal phase path rather than assuming global differentiability.*

Proof. Additivity and compactness give dynamic programming. The finite projected pressure/rate identities commute with endpoint contractions by Theorem 1.6. Projective limits are justified by Lemma 1.3. Local Gaussian tangents are Theorem 1.7. Taking minima or maxima over the finitely active phase charts preserves the stated identities. $\square$

Theorem 1.9 (Round-seven D1 closure). *The projective theorem uses local zero-free complex atlases, bounded-support face stratification, phase-wise recovery at coexistence, an explicit topology upgrade, genuine thin-shell conditioning, complete finite-dimensional duality hypotheses, and exact finite-centred local likelihoods. It requires neither an entire complex log-mgf nor a gradient diverging at infinity and provides a positive standalone stratified projective commutation theorem.*

Proof. Combine Lemmas 1.1 and 1.3 with Theorems 1.2, 1.4, 1.5, 1.6, 1.7, and 1.8. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are

necessary verification gates; independent external journal review remains a separate process.

PEKING UNIVERSITY, BEIJING 100871, CHINA

Email address: `qiqian@pku.edu.cn`